AI HR Interview System - Technical Documentation

Audit Report: 33 Signals & 33 Core Modules

1. Proctoring & Behavioral Signal Matrix (33 Signals)

#	Signal Name	Status	Implementation Tools
1	Tab Switch	Ready	LiveKit / JS
2	Face Absent	Ready	MediaPipe / LiveKit
3	Multiple Faces	Ready	OpenCV / LiveKit
4	Gaze Violation	In-Progress	FaceMesh AI
5	Ownership Ambiguity	Ready	Python (NLP)
6	Example Evasion	Ready	OpenAI (GPT-4o)
7	Follow-up Collapse	Ready	Custom Heuristics
8	Latency Mismatch	Ready	Deepgram Timing
9	Factual Contradiction	Ready	OpenAI (GPT-4o)
10	AI-Read Pattern	Ready	NLP + Timing
11	Lookup Delay	Ready	NLP + Timing
12	Resume Mismatch	In-Progress	Pinecone / OpenAI
13	Depth Decay (Mild)	Ready	Statistical Model
14	Depth Decay (Severe)	Ready	Statistical Model
15	Meta-Questioning	In-Progress	Regex Matching
16	Signal Surfaced	Ready	State Engine
17	Signal Unresolved	Ready	State Engine
18	Signal Resolved	Ready	State Engine
19	Signal Repeated	Ready	State Engine
20	Deep Logic Score	Ready	OpenAI (GPT-4o)
21	Specificity Score	Ready	OpenAI (GPT-4o)
22	Surface Score	Ready	OpenAI (GPT-4o)
23	Critical Surface	Ready	OpenAI (GPT-4o)
24	Ownership Recovery	Ready	NLP Rules
25	Example Recovery	Ready	NLP Rules
26	Pressure Stability	Ready	Custom Math
27	Improving Trend	Ready	PostgreSQL Logic
28	Declining Trend	Ready	PostgreSQL Logic
29	Collapsing Trend	Ready	PostgreSQL Logic
30	Clarity Tracker	Ready	Behavioral Core
31	Scripted Pattern	In-Progress	OpenAI (GPT-4o)
32	Confidence Gap	Ready	Behavioral Core
33	Evasion Cluster	Ready	Behavioral Core

2. Module Audit (33 Core Python Files)

#	File Name	Role	Real-Time Example
1	agent.py	Worker Entry Point	Initializes the LiveKit connection for a candidate.
2	agent_runtime.py	Session Orchestrator	Manages model prewarming and session lifecycle.
3	trust_engine.py	Trust Modeler	Applies mathematical deltas when red flags are detected.
4	turn_decision.py	Signal Detection	Runs regex-based checks for Ownership Ambiguity.
5	agent_persistence.py	Data & Billing	Tracks token usage and saves transcript to PostgreSQL.
6	agent_scoring.py	LLM Scorer	Gets 0-10 scores from OpenAI on Depth and Specificity.
7	behavioral_analysis.py	Narrative Engine	Summarizes interview history into recruiter insights.
8	probe_strategy.py	AI Follow-up Tactics	Tells AI which 'Probe' to use for vague answers.

9	prompt_layers.py	Instruction Manager	Defines the AI's personality and evaluation rules.
10	agent_profile_context.py	Candidate Context	Tells AI to focus on seniority-based questions.
11	agent_providers.py	API Bridge	Connects to Deepgram, Cartesia, and OpenAI.
12	agent_runtime_helpers.py	Turn Management	Ensures AI asks only one question at a time.
13	agent_runtime_support.py	Logging & Config	Generates structured JSON logs for debugging.
14	agent_utils.py	Data Sanitizer	Cleans messy text from resumes or transcripts.
15	contradiction_state.py	Logic Sync	Tracks if an inconsistency has been surfaced.
16	recruiter_summary.py	Report Generator	Combines all scores into a final recommendation.
17	session_state.py	State Storage	Holds live Trust Score for turn-by-turn logic.
18	convert_raw.py	Log Processor	Converts raw logs into readable transcripts.
19	remove_empty_lines.py	File Cleanup	Optimizes transcript file size by removing whitespace.
20	extract_pdf.py	Resume Parser	Extracts text from PDF resumes for AI analysis.
21	audit_tags.py	DB Auditor	Ensures interviews have correct job/org tags.
22	fix_20_21.py	Data Fixer	Corrects historical date errors in old records.
23	fix_agent.py	Sandbox Agent	Testing version of AI for model trials.
24	fix_phone_inputs.py	Contact Sanitizer	Formats messy phone numbers into global E.164.
25	fix_settings_async.py	Settings Patch	Ensures admin settings updates are saved instantly.
26	fix_transcript.py	Recovery Tool	Re-generates transcript from recorded audio.
27	fix_validation_errors.py	DB Cleaner	Repairs broken database records.
28	remove_bom.py	Text Cleaner	Removes invisible characters from resume files.
29	repair_dashboard.py	Search Re-indexer	Fixes search results on the admin dashboard.
30	total_recovery.py	System Restorer	Re-links interviews after database crashes.
31	fix_models.py	DB Schema Tool	Adds new data columns like 'Cheating Score'.
32	test_agent_helpers.py	Logic Tester	Runs automated brain-logic tests.
33	requirements.txt	Dependency List	Lists all libraries needed to run the system.